

Week 2 Lab – CS010A

Programming a Chemistry Formula

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Before We Start: A Few Updates

Lab 1 grades

- ▶ We are working on getting the student rosters organized per TA.
- ▶ Because of this, Lab 1 grading is delayed.
- ▶ Apologies for the delay.

Lab sheets

- ▶ Starting today: **one sheet per group**, rather than one per student.

TA Office Hours

- ▶ Thursday: **5:00–6:00 PM**, Zoom
- ▶ Friday: **2:00–3:00 PM**, in-person, **WCH 110**
- ▶ Full details are in the Canvas announcement.



Why Program Mathematical Equations?

Computers are very good at performing the same calculation **quickly and repeatedly**.

In science and engineering, programs are commonly used to:

- ▶ evaluate mathematical models,
- ▶ process experimental measurements,
- ▶ run simulations,
- ▶ test many different input values.

Today we will take a mathematical equation and **translate it into a C++ program**.

Important

You do **not** need to know chemistry or physics for this lab.
The equation and constants are already given.



Today's Equation: Van der Waals Equation

The Van der Waals equation relates pressure, volume, temperature, and the amount of gas:

$$\left(P + a\frac{n^2}{V^2}\right)(V - nb) = nRT$$

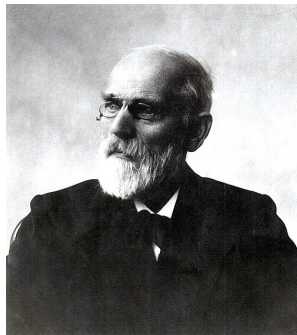
For today's lab:

- ▶ Inputs: P , V , n
- ▶ Output: T
- ▶ $a = 3.59$
- ▶ $b = 0.0427$
- ▶ $R = 8.314$

Your first task

Use simple algebra to rewrite the equation so that T is by itself.

Do not unnecessarily simplify the equation.



Johannes Diderik van der Waals



C++ Tools You Need Today

For this lab, use **float**:

```
float P, V, n;  
float T;  
  
float a = 3.59;  
float b = 0.0427;  
float R = 8.314;
```

Take input using **cin**:

```
cin >> P >> V >> n;
```

Basic arithmetic:

+ - * /

Remember

Parentheses matter.

Try to make your C++ expression resemble the mathematical equation as closely as possible.



Powers and the cmath Library

C++ provides the `cmath` library for additional mathematical operations:

```
#include <cmath>
```

For example:

```
pow(x, 2)
```

means

$$x^2$$

Example:

```
float x = 3.0;  
float result = pow(x, 2);
```

So result becomes 9.

For today's equation

Look carefully for quantities that are **squared**.



More mathematical powers!

Functions such as `pow()`, `sqrt()`,
etc.



Break Large Calculations Into Smaller Steps

You do **not** need to write the entire equation as one giant C++ expression.

Use temporary variables:

```
float part1 = ...;
float part2 = ...;

float result = part1 * part2;
```

Why?

- ▶ easier to read,
- ▶ easier to compare with the equation,
- ▶ easier to find mistakes,
- ▶ easier to debug.



Separate pieces Each part has a clear role.



Everything mixed
Harder to inspect when something goes wrong.

Good programming habit

Break complicated calculations into **small logical pieces**.

Before You Write C++

Follow the same process as last week:

1. Read the problem specification carefully.
2. Write your **code plan** in plain English.
3. Write your **test cases** and expected outputs.
4. Then translate the plan into C++.

Provided reference test

Expected output

Input

100 10 1

119.808

Do not skip the planning

Complete the **code plan** and **test cases** before writing C++.



Lab 2: Chemistry Formula

The Van der Waals equation is

$$\left(P + a \frac{n^2}{V^2}\right) (V - nb) = nRT$$

Write a program that:

- ▶ takes P , V , and n as inputs,
- ▶ computes T ,
- ▶ uses $a = 3.59$, $b = 0.0427$, $R = 8.314$,
- ▶ declares all variables as `float`,
- ▶ assumes all inputs are positive.

Hint: Temporary variables may help you store partial calculations.

Start Working

Groups of 2

Read Specs → Code Plan → Test Cases → C++ Code



Images references

- ▶ Johannes van der Waals:
https://en.wikipedia.org/wiki/Johannes_Diderik_van_der_Waals
- ▶ Confused Thanos:
https://www.reddit.com/r/dankmemes/comments/bxdzic/confused_thanos/
- ▶ Gauntlet: <https://www.reddit.com/r/marvelstudios/s/zFDaRUwAit>
- ▶ Prismatic stone: <https://www.ebay.com/itm/134278030931>
- ▶ Power stone: <https://www.reddit.com/r/UnearthedArcana/s/Be7zvHhzXd>

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